



Context-Aware NL2SPARQL with Interactive Entity Linking

KG Lab WS2025 | Group 2

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Valuable Data Trapped in Knowledge Graphs

- Knowledge graphs store billions of RDF triples across research, business, and government domains
 - DBLP alone consists of 548 million RDF triples
- But accessing this data requires SPARQL expertise
- **Result:** Valuable insights remain locked away behind technical barriers

LLMs offer a promising bridge for natural language interfaces

Solution: Natural Language to SPARQL with user in the loop

Domain-Agnostic Pipeline

1. **Context Pack:** Load in needed data (SPARQL endpoint, KG schema, example queries)
2. **Mention Extraction:** LLM identifies entities in the user question
3. **Entity Linking:** Find matching entities in the KG and rank them
4. **Interactive Approval:** User verifies the correct entities
5. **Query generation:** Guided by schema exploration tools, examples, and linked entities
6. **Execution:** Run validated query on provided endpoint

Context Pack

- Required Input:
 - Schema/Ontology in any serialization
 - SPARQL endpoint
 - base IRI
- Optional Input:
 - Examples
 - Description
- We need to know: RDF, RDFS, OWL:
 - Classes
 - Hierarchies
 - Domain and range of properties
 - Labels and comments
 - etc.
- Turn schema file into 3 dictionaries for
 - namespaces (prefixes)
 - classes
 - properties

Schema Name

DBLP

Endpoint URL

https://sparql.dblp.org/sparql

Base IRI

https://dblp.org/rdf/schema#

Description

The DBLP computer science bibliography. Contains data about publications, authors, venues, etc.

95 / 500

Examples

+ Add Example

Question

Who are the authors of "Attention Is All You Need"?

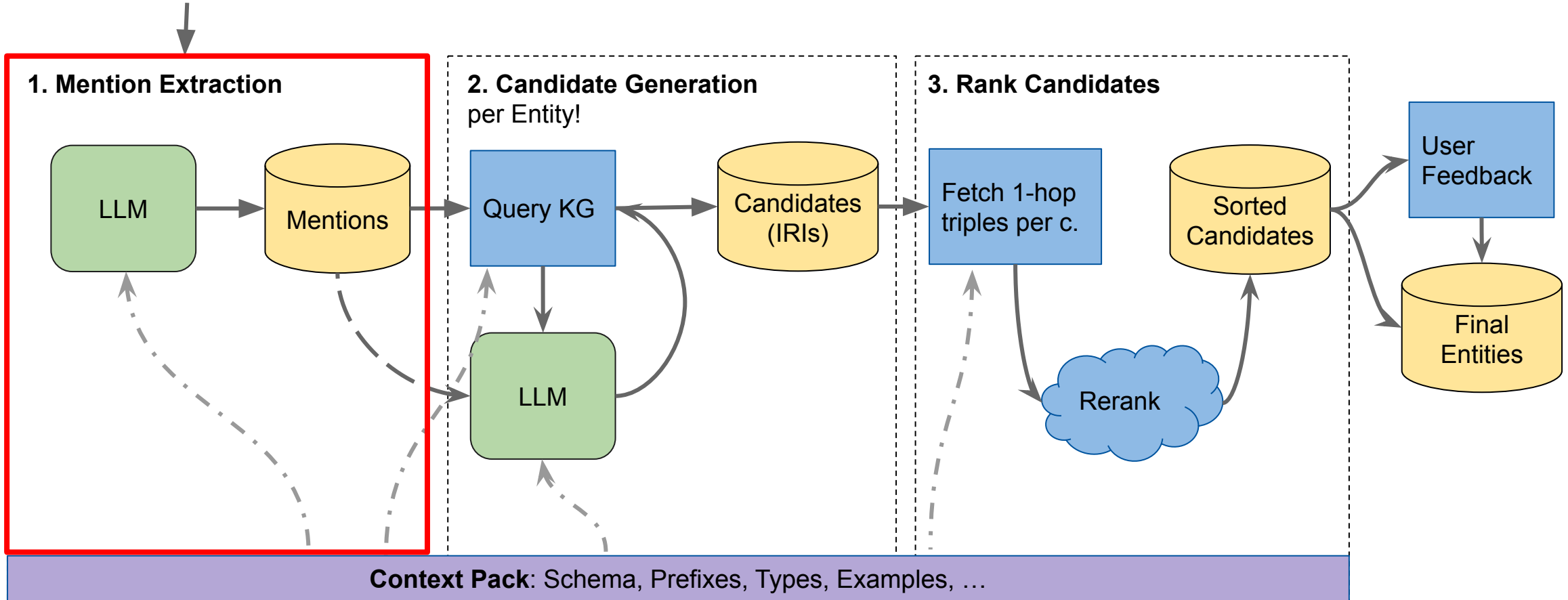
SPARQL

```
1 PREFIX dblp: <https://dblp.org/rdf/schema#>
2 SELECT ?author ?authorName WHERE {
3   BIND (<https://dblp.org/rec/journals/corr/VaswaniSPUJGKP17> AS ?pub)
4   ?pub dblp:authoredBy ?author .
5   OPTIONAL { ?author dblp:creatorName ?authorName . }
6 }
```

Remove

Entity Linking Pipeline

Query: Who are the co-authors of Vaswani in Attention Is All You Need?



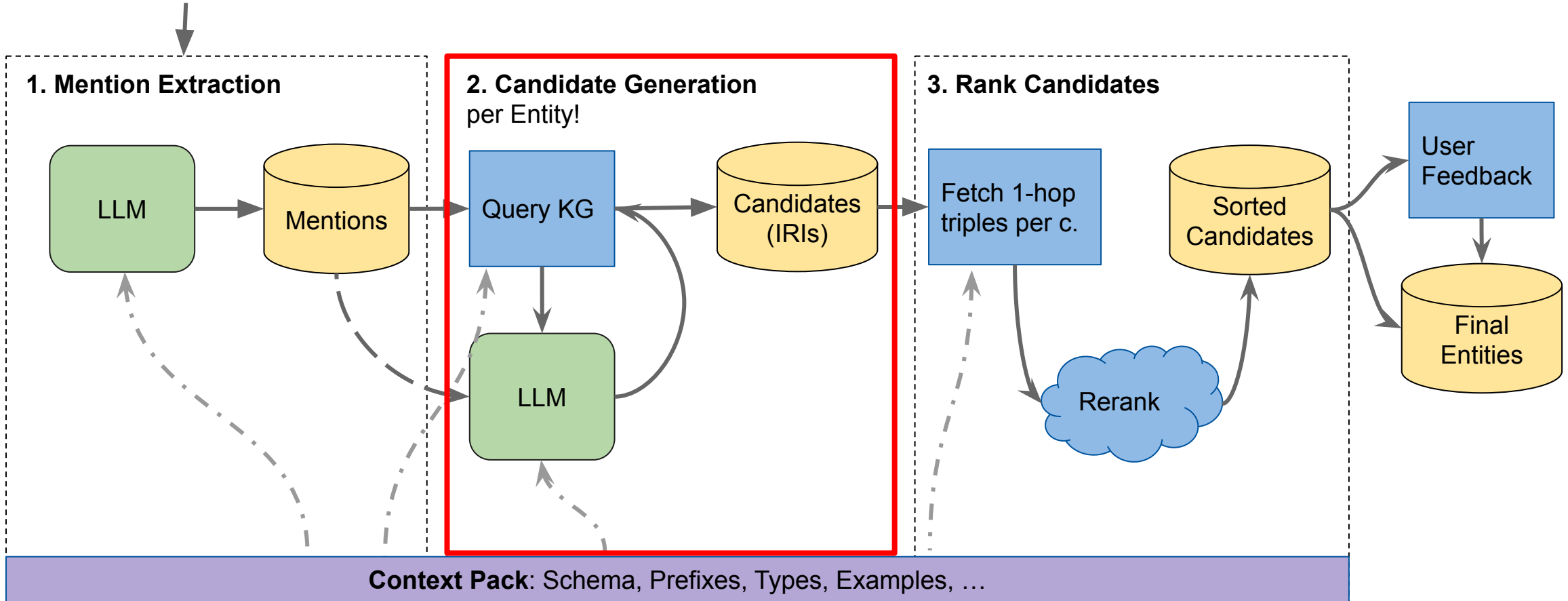
Who are the co-authors of Vaswani in Attention Is All You Need?

```
Vaswani -  
  type - dblp:Creator  
  label_pred - dblp:creatorName
```

```
Attention Is All You Need -  
  type - dblp:Publication  
  label_pred - dblp:title
```

Entity Linking Pipeline

Query: Who are the co-authors of Vaswani in Attention Is All You Need?



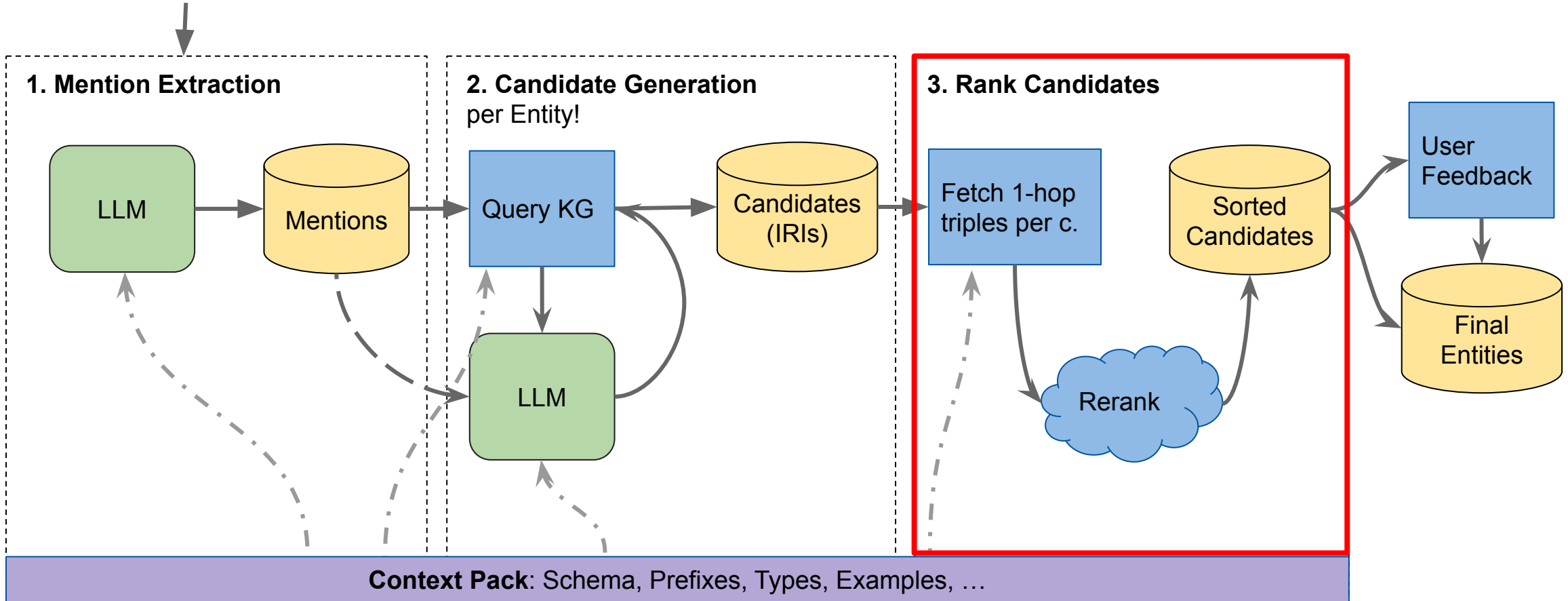
Candidate Generation

- Candidate Retrieval Pipeline
 - Strict Search (Fast & Precise): A fast SPARQL query looking for an exact substring match.
 - Handles standard entities perfectly.
- Relaxed Token Search (Robust Fallback)
 - If strict search fails (often due to "First Last" vs "Last First" name variations), we tokenize the mention and run an OR query across the tokens.
 - Database-Level Prioritization: To prevent relaxed queries from returning too much noise, the SPARQL query natively sorts results before applying a LIMIT 30. It prioritizes:
 - Exact Matches
 - Highest number of matched tokens
 - Node Degree (popularity)
 - Shortest label length
 - Lexicographically

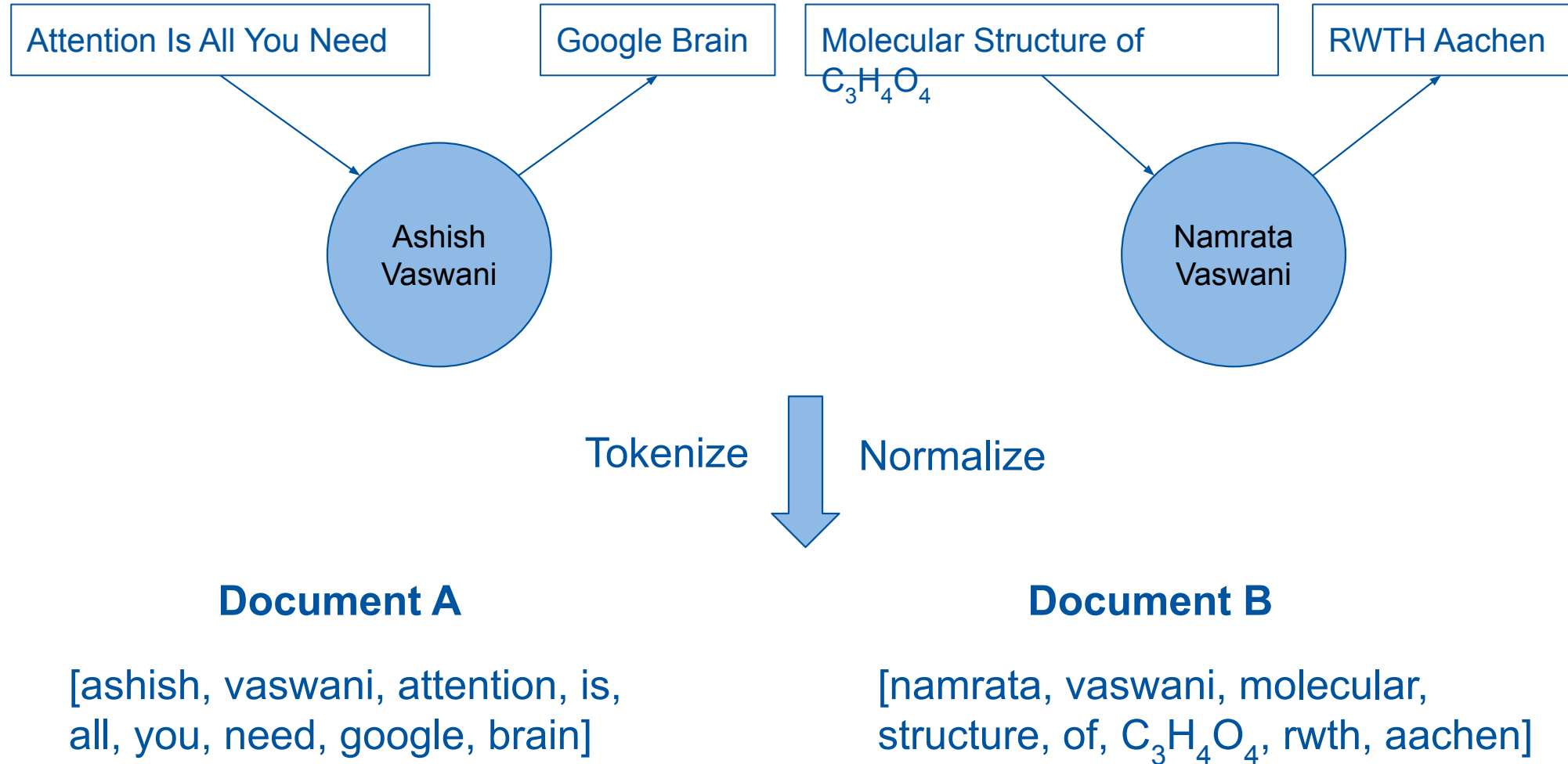
} Used for boosting ranking

Entity Linking Pipeline

Query: Who are the co-authors of Vaswani in Attention Is All You Need?



Ranking - Document Generation



Ranking - Intuition

[ashish, vaswani, attention, is,
all, you, need, google, brain]

Document A

[namrata, vaswani, molecular,
structure, of, C₃H₄O₄, rwth, aachen]

Document B

Query

[vaswani, attention, is, all, you, need]

Hybrid Scoring Function: BM25 + Candidate generation scoring factors

Document	Score
Document A	175
Document B	55

Query: Who are the co-authors of Vaswani in Attention Is All You Need?

1. Extracted Mentions

Extracted Mentions:

Publication: Attention Is All You Need

—

Creator: Vaswani

2. Candidate Generation

- Found 4 candidates, 2 unique IRIs (Publication)
- Found 21 candidates, 21 unique IRIs (Person)

3. Ranked Candidates

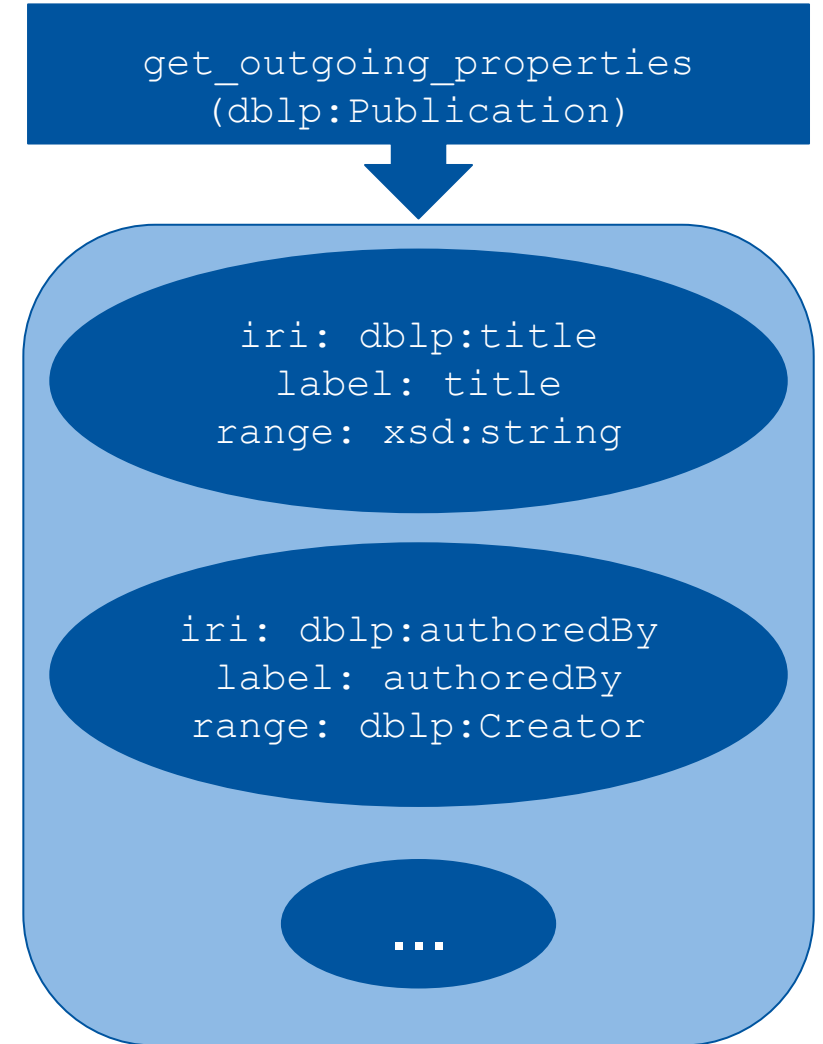
- Score: 175.23
iri=<https://dblp.org/rec/journals/corr/VaswaniSPUJGKP17>,
label=Ashish Vaswani et al.: Attention Is All You Need. (2017)
- Score: 172.84
iri=<https://dblp.org/rec/conf/nips/VaswaniSPUJGKP17>,
label=Ashish Vaswani et al.: Attention is All you Need. (2017)
-
- Score: 120.57
iri=<https://dblp.org/pid/26/9012>, label=Ashish Vaswani
- Score: 57.13
iri=<https://dblp.org/pid/270/9442>, label=Namrata Vaswani
- Score: 4.55
iri=<https://dblp.org/pid/284/7374>, label=Chirag Vaswani Bhavnani

Query Generation - Prompt Construction

- LLM requires additional information about the KG
- Prompt includes:
 - System prompt (“You are a SPARQL query generator...”)
 - Question/query examples
 - User question
 - Linked entities
 - Tool box
- Send prompt to LLM which then starts the tool-calling loop

Query Generation - Tool calling loop

- List of tools
 - `list_classes()`
 - `describe_class(classIRI)`
 - `describe_property(classIRI)`
 - `get_outgoing_properties(classIRI)`
 - `get_incoming_properties(classIRI)`
 - `run_sparql_query(query)`
- Traverses schema graph using `get_outgoing_properties` and `get_incoming_properties`
- Uses `run_sparql_query` at the end to validate and run the query



Query Validation

- 4 Validation Steps
 1. Check basic syntax of SPARQL query
 2. Check that query type is: SELECT, ASK
 3. Check that all used prefixes are in the schema index
 4. Check that all properties and “type objects” of the base IRI are present in the schema index
- If validation process unsuccessful → display what went wrong

```
1 PREFIX dblp: <https://dblp.org/rdf/schema#>
2 SELECT DISTINCT ?coauthor ?coauthorName WHERE {
3   BIND (<https://dblp.org/pid/56/953> AS ?author)
4   ?pub a dblp:Publication ;
5       dblp:authoredBy ?author ;
6       dblp:writtenBy ?coauthor .
7   FILTER(?coauthor != ?author)
8   OPTIONAL { ?coauthor dblp:creatorName ?coauthorName . }
9 }
```

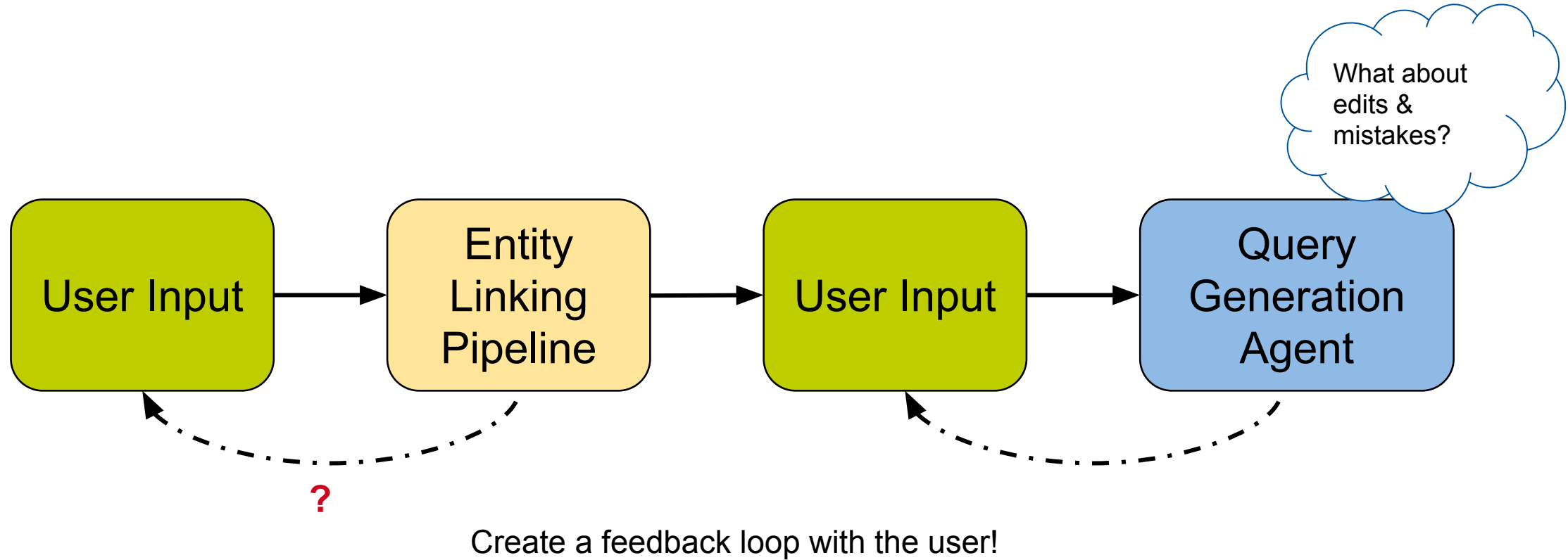
Term <https://dblp.org/rdf/schema#writtenBy> not found in schema.

Evaluation - Entity Linking

- Tested on 100 questions from DBLP-QuAD: multiple query types, adapted venues: string → IRI
- Used `gpt-oss:120b`
- **Evaluation Logic:**
 - Per question: exact mention spans, exact IRI matches
 - Per entity: Top1 Rate, Recall@10
- **Results:**
 - Mentions: F1=0.99 (it extracted the title which was deemed to be incorrect)
 - **Entity Linking F1=0.83**
 - Top1 Rate = 0.85; Recall@10 = 0.89
- **Limitations:**
 - Ambiguous abbreviated author names “Alberto R” are often underspecified
 - Substring bias (CONTAINS) - B. Anda → Kristine Claire B. Andaya
 - No full text/semantic search.
 - Linking quality depends on schema fidelity

Evaluation - Query Generation

- Tested on the same dataset as Entity Linking
- Problems with DBLP-QuAD (venues as IRIs, invalid queries)
- We used `gpt-oss:120b`
- **Evaluation Logic:**
 - ASK queries: check if boolean matches
 - SELECT queries:
 - check solutions of dataset are present in the solutions of the generated query
 - check that number of solution mappings matches
 - idea: do not punish if additional variables were returned
- **Results: 86 out of 100 were correct**
- **Limitations:**
 - Performance degrades when the ontology does not match the actual dataset (e.g. `dblp:authorOf`)
 - Reasoning and context capabilities of the LLM → We expect this to be better in the future.



Chat Agent

How many papers has Andrew Ng published in NeurIPS?

User

Oh sorry it is not Andrew Ng, it should be Andrew Y Ng.

Can you group the results by year and add title of the publications?

Routing Agent

Entity Linking Agent

Query Generation Agent

State = List[Mentions]
User Input + Feedback
Tools:

- State Operations: List, Edit, Add, Delete
- Schema Navigation
- Generate Candidates
- Rank Candidates
- Inspect & Boost

State = Current Query
User Input + Feedback
Tools:

- Schema Navigation
- Run Query

Evaluation Summary (Linking-agent)

Out of the box! No prompt optimization of any kind.

Total Queries : 100
Extraction F1 : 0.99
Linking F1 : 0.84
Entity Recall@10 : 0.90
Entity Top-1 Rate : 0.85

Noticed

- Similar score as before but different errors!
- Agent has more flexibility at ranking and choosing candidates.
 - Choosing ranking keywords
 - Boosting candidates

```
Tool call name=create_mention args={'text': 'IEEE Wirel. Commun.', 'type_curie': 'dblp:Journal', 'label_p  
Tool call name=search_candidates args={'mention_id': 1}  
Tool call name=rerank_mention args={'mention_id': 1, 'keywords': ['IEEE', 'Wireless', 'Communications']}
```

Future Work

- **End-to-End Evaluation**
 - Evaluate the full agentic NL2SPARQL pipeline (extraction → linking → query → validation → execution) and measure accuracy, latency, and cost.
- **Agent Design**
 - Improve model-tool interaction by using skills for better retrieval, re-ranking, and iterative search-and-repair.
- **Prompt & Interaction Design**
 - More guided clarification and optimize prompts for different query types.
- **Cross-KG Validation**
 - Test transferability beyond DBLP.

Thank you for listening. Questions?



- Context Pack
- Entity Linking Pipeline
 - Mention Extraction
 - Candidate Generation
 - Ranking Candidates
- Query Generation
 - Prompt Construction
 - Tool calling loop
- Query Validation Steps
- Evaluation

Appendix

All slides after this are a part of the appendix

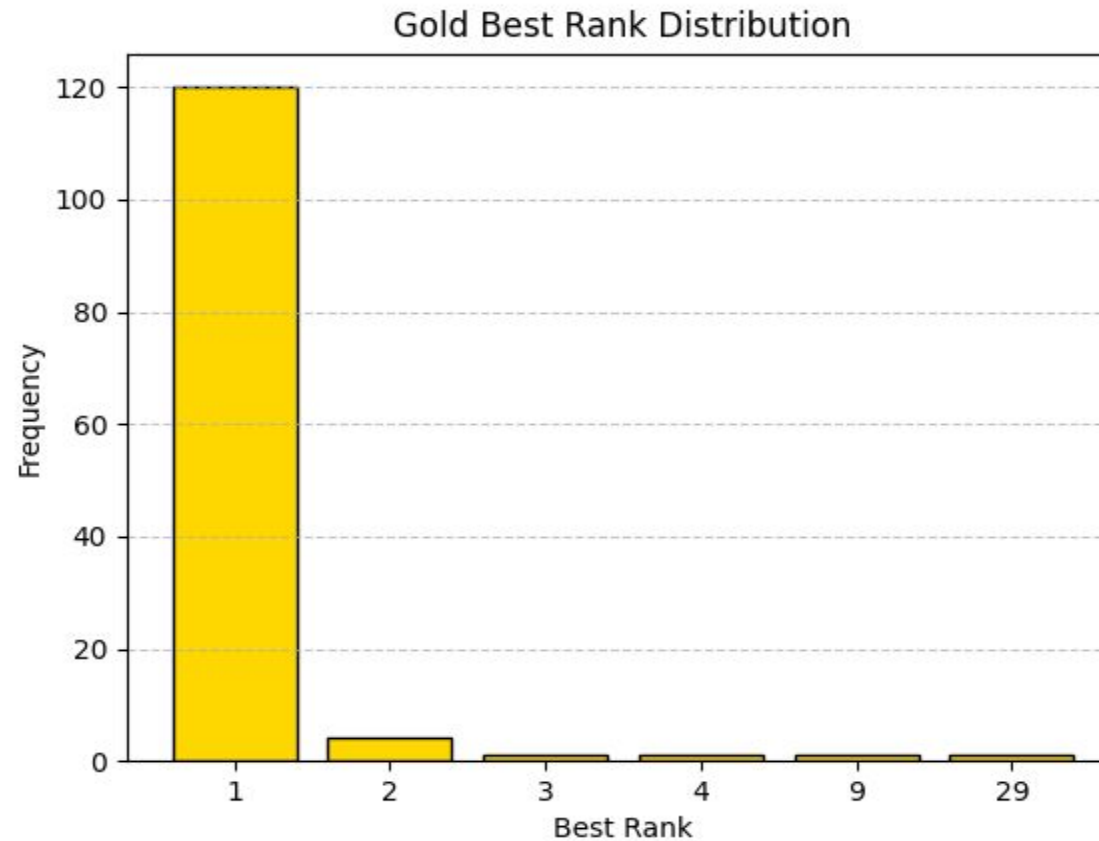
Current Solutions

- **Domain-Specific Systems** (Too Rigid):
 - DBLPLink 1.0 / 2.0: Specific trained models, prompts and APIs.
 - ASK-DBLP: Tightly coupled to DBLP schema, relies on DBLPLink 2.0.
- **Generic LLM Systems** (Too Loose):
 - ChatGPT/Claude: Hallucinate properties, wrong entities, low effort produces bad results
 - No validation: Queries may be syntactically invalid
 - A lot of user back and forth to get the right data
- **Agnostic Systems**
 - sparql-LLM: Generates queries over multiple KGs in the domain of bioinformatics
 - NL2SPARQL (our system) 🎉

Mention Extraction Prompt

```
1 You are an entity mention extractor for a knowledge graph.
2
3 Goal
4 Extract entity mentions from the user query and select the correct STRING-LABEL predicate
5 to later retrieve matching entities from the knowledge graph.
6
7 Hard rules
8
9 - Only output spans that are contiguous substrings of the user text. NEVER invent text.
10 - Every mention text and every attr value MUST appear verbatim in the user query.
11 - Do not infer or complete missing information.
12 - Do not output duplicate (text, type) pairs.
13 - Return ONLY valid JSON.
14
15 How to choose `type`
16
17 - `type` MUST be exactly one of the Types listed in Schema Context.
18 - Output the TYPE CURIE exactly as shown (the left of the "|"), NOT the label text.
19
20 How to choose `label_pred` (CRITICAL)
21
22 - `label_pred` MUST be exactly one predicate CURIE listed under the chosen `type`.
23 - IMPORTANT: In Schema Context predicate rows have the shape:
24 |   pred_curie | pred_label | rng:...
25 | You MUST output the pred_curie (left side). Ignore pred_label entirely.
26 - `label_pred` MUST have rng:lit(xsd:string).
27 - **Identity vs. Attribute**: Choose the predicate that stores the **Name, Label, or Identifier** of the entity itself.
28 - **CRITICAL**: Do NOT select a predicate just because its label appears in the user query.
29 - If the user asks for "the *population* of **Paris**", the mention is "Paris" and the `label_pred` should be a name/label predicate (e.g., `rdfs:label`), NOT the "population" predicate.
30 - The `label_pred` is used to *find* the entity by the `text` you extracted. The attributes the user is asking *about* will be handled in a later step.
31 - Choose the predicate whose VALUES are most likely to contain the mention text verbatim.
32 - If multiple predicates are plausible, pick the best one.
33 - **CRITICAL**: You must extract the entity exactly as it appears in the source text, character-for-character. Do not alter capitalization, accents, or punctuation. However, if the entity is
    enclosed in quotation marks that act as delimiters in the query (e.g., 'Paris' or "Nature"), do NOT include those surrounding quotes in the extraction.
34
35 Attrs
36
37 - attrs are optional metadata, NOT part of the mention text.
38 - Only use attribute keys that appear in Schema Context for that type.
39 - Values MUST be verbatim substrings of the user query.
40 - Do not emit empty strings.
41 - Year ranges:
42 | If a range appears (e.g., "2015-2020" or "between 2015 and 2020"),
43 | store it on ONE mention as "year_start":"2015","year_end":"2020".
44
45 Schema Context
46 {string_ctx}
```

Entity Linking - Histogram



Out of 142 entities, 14 entities were not retrieved. Out of 128 entities retrieved, 120 were the top choice.

Candidate Generation Queries

```
1 SELECT ?s ?p ?label
2   (STRLEN(STR(?label)) AS ?labelLen)
3   (IF(LCASE(REPLACE(STR(?label), "\\.$", ""))=LCASE("Vaswani"), 1, 0) AS ?exact)
4   (COUNT(?extra_o) AS ?degree)
5 WHERE {
6   VALUES ?cls { <https://dblp.org/rdf/schema#Creator> }
7   VALUES ?p { <https://dblp.org/rdf/schema#creatorName> }
8   ?s a ?cls ; ?p ?label .
9   FILTER(isLiteral(?label))
10  FILTER(CONTAINS(LCASE(STR(?label)), LCASE("Vaswani")))
11  FILTER(LANG(?label) = "" || LANGMATCHES(LANG(?label), "en"))
12  OPTIONAL { ?s ?extra_p ?extra_o }
13 }
14 GROUP BY ?s ?p ?label
15 ORDER BY DESC(?exact) DESC(?degree) ASC(?labelLen) LCASE(STR(?label))
16 LIMIT 30
```

BM25 + Candidate Generation Scoring Factors

$$\text{Score}_{BM25} = \sum_{t \in Q} \text{IDF}(t) \cdot \frac{f(t, D) \cdot (k_1 + 1)}{f(t, D) + k_1 \cdot (1 - b + b \cdot \frac{|D|}{\text{avgdl}})}$$
$$\text{IDF}(t) = \ln\left(1 + \frac{N - n_t + 0.5}{n_t + 0.5}\right)$$

where $f(t, D)$ is the number of times that the keyword t occurs in the document D , $|D|$ is the length of the document D in words, and avgdl is the average document length in the text collection from which documents are drawn.

k_1 and b are free parameters, usually chosen, in absence of an advanced optimization: $k_1 = 1.5$ and $b = 0.75$ in our solution

- For exact match, we add +100 to the score
- For each token match, we add +50 to the score
- A small $(\log_{10}(\text{degree} + 1.0) * 0.1)$ score is added based on node degree

Query Generation Prompt

You are a SPARQL query generator.

You have access to schema exploration tools to discover:

- Classes
- Properties
- Domains and ranges

Guidelines

- Do NOT assume class names or property names
- Do NOT hallucinate 🐱 predicates
- Use the extracted mentions and their linked IRIs
- When doing SELECT queries, when returning a subject or object, always return the IRI in addition to results requested by the user.
- Only use string-based filtering as a last resort when no IRI is available for a mention. In that case prefer ``CONTAINS(LCASE(?var), LCASE("..."))`` over exact literal equality.
- Use schema exploration tools before generating the SPARQL query
- ALWAYS run the query with ``run_sparql_query`` to validate the syntax and verify endpoint execution behavior before returning it.
 - If ``run_sparql_query`` reports timeout, simplify the query and try again (fewer joins/filters, add stricter constraints, add reasonable LIMIT).
 - If ``run_sparql_query`` reports syntax/forbidden/endpoint error, revise the query before returning.
- Always include a PREFIX declaration section at the top of the query using standard abbreviations (e.g., `rdfs:`, `xsd:`, `dblp:`) to ensure the SELECT clause remains concise and readable.
- Return ONLY the SPARQL query
- Do NOT add comments or explanations

Examples

{examples}

Tool calling Loop Example

Return all papers (IRI and title) of Nicholas Carlini

- Starts with linked entity: Nicholas Carlini, type:Creator

1. list_classes()
2. describe_class(dblp:Publication)
3. get_outgoing_properties(dblp:Publication)
4. run_sparql_query(query)

```
1  PREFIX dblp: <https://dblp.org/rdf/schema#>
2  SELECT DISTINCT ?paper ?title WHERE {
3      BIND (<https://dblp.org/pid/145/1806> AS ?creator)
4      ?paper a dblp:Publication ;
5              dblp:authoredBy ?creator ;
6              dblp:title ?title .
7  }
```

High-Level Architecture

The system follows a client-server architecture separating the UI from the heavy lifting of model inference.

Frontend (Client)

- **Framework:** Next.js 16 (React 19)
- **Role:** User interaction, visualization, State management, local storage
- **Communication:** REST API calls to Backend.

Backend (Server)

- **Language:** Python + RDFLib
- **Role:** Mention extraction, prompt construction, SPARQL validation, SPARQL execution.
- **AI Engine:** Ollama (RWTH Endpoint inference).

Frontend Step 1 - Question Input

List all coauthors of Yoshua Bengio.

☐

Skip Mention Extraction

SCHEMA: DBLP ▾

MODEL: RWTH-GPT-gpt-oss-120b ▾



TRY:

Who are the authors of the paper named: Attention is All You Need?

List all coauthors of Yoshua Bengio.

Which papers did Dan O. Popa publish in the last 9 years?

What is the average number of papers published by Darko Kirovski per year?

What is the webpage of Ravi Kumar?

^ Detected Mentions

! Instructions

Click the "Show more" button to expand each mention, then select the best matching candidate from the list.



A Priority Queue Transform

1 entity candidate

∨ Show More

✕ Remove

Mention Type

<https://dblp.org/rdf/schema#Publication>

Best Match

A Priority Queue Transform.

<https://dblp.org/rec/conf/wae/Fredman99>

Score: **103.4978**



Michael L. Fredman

1 entity candidate

∨ Show More

✕ Remove

Mention Type

<https://dblp.org/rdf/schema#Creator>

Best Match

Frontend Step 2 - Entity Linking

Best Match

Ashish Vaswani et al.: Attention is All you Need. (2017)

<https://dblp.org/rec/conf/nips/VaswaniSPUJGKP17>

Score: **98**

Other Candidates

Hongliu Wu et al.: Not All Attention Is All You Need. (2021)

<https://dblp.org/rec/journals/corr/abs-2104-04692>

Score: **85**

Ashish Vaswani et al.: Attention is All you Need. (2017)

<https://dblp.org/rec/conf/nips/VaswaniSPUJGKP17>

Score: **98**

Ashish Vaswani et al.: Attention Is All You Need. (2017)

<https://dblp.org/rec/journals/corr/VaswaniSPUJGKP17>

Score: **96**

Frontend Step 3 - Edit SPARQL

Edit SPARQL Query

```

1 PREFIX dblp: <https://dblp.org/rdf/schema#>
2 PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
3
4 SELECT DISTINCT ?coauthor ?coauthorName ?publication WHERE {
5     # Find the publication "Attention is All You Need"
6     ?publication rdfs:label "Attention is All you Need. (2017)" ;
7     |         |         |         |         |         |
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← Back

Execute Query

Frontend Step 4 - Show Query result

Query Results

Found 7 results

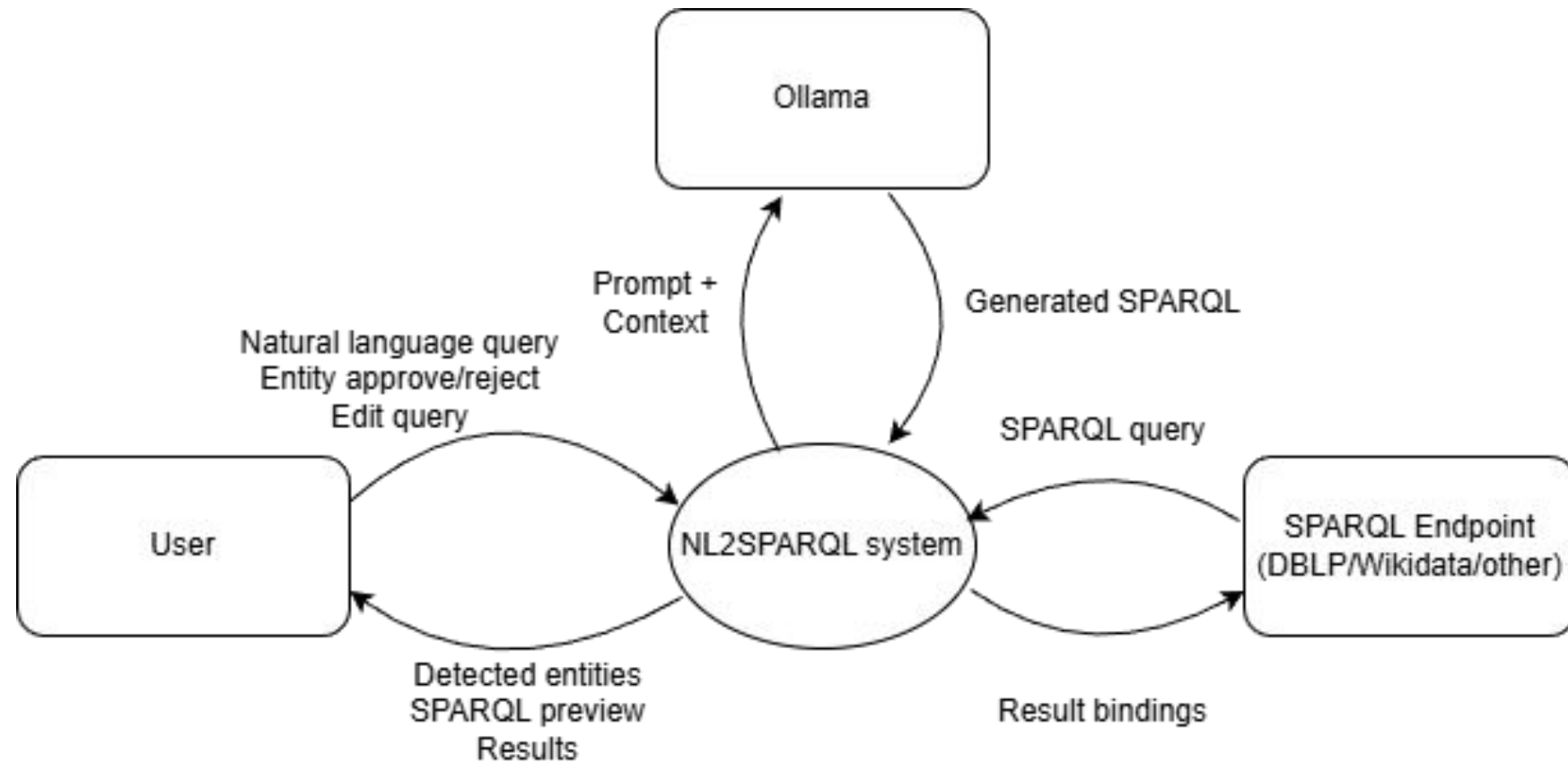
Results

Label	URI	Type
Illia Polosukhin	https://dblp.org/pid/p/IlliaPolosukhin	https://dblp.org/rdf/schema#Person
Noam Shazeer	https://dblp.org/pid/s/NoamShazeer	https://dblp.org/rdf/schema#Person
Parmar Niki	https://dblp.org/pid/p/ParmerNiki	https://dblp.org/rdf/schema#Person
Jakob Uszkoreit	https://dblp.org/pid/u/JakobUszkoreit	https://dblp.org/rdf/schema#Person
Llion Jones	https://dblp.org/pid/j/LlionJones	https://dblp.org/rdf/schema#Person
Aidan N. Gomez	https://dblp.org/pid/g/AidanGomez	https://dblp.org/rdf/schema#Person
Lukasz Kaiser	https://dblp.org/pid/k/LukaszKaiser	https://dblp.org/rdf/schema#Person

References

- DBLPLink: An Entity Linker for the DBLP Scholarly Knowledge Graph, Debayan Banerjee et al., 2023
- DBLPLink 2.0 - An Entity Linker for the DBLP Scholarly Knowledge Graph, Debayan Banerjee et al., 2025
- sparql-LLM: LLM-based SPARQL Query Generation from Natural Language over Federated Knowledge Graphs, Vincent Emonet et al., 2024
- ASK-DBLP: Answering Questions over DBLP, Tilahun Abedissa Taffa et al., 2025
- Okapi BM25: https://en.wikipedia.org/wiki/Okapi_BM25

System Context Diagram



UML Use Case Diagram

